FleetEase: A Simulation-Validated ADAS Framework for Heavy Commercial Vehicles in Indian Urban Traffic Scenarios

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Abstract

Heavy commercial vehicles (HCVs) account for a disproportionately large share of fatal crashes on Indian roads relative to their share of the registered vehicle population. Mainstream Advanced Driver Assistance System (ADAS) stacks have been developed and calibrated for passenger cars in lane-disciplined Western traffic, which substantially limits their transferability to Indian heavy-duty operation. This work introduces **FleetEase**, a Level-2 ADAS framework whose perception and decision stages are validated entirely in simulation prior to on-vehicle deployment. The framework couples the CARLA open urban driving simulator with road geometry authored in MathWorks RoadRunner, producing a test substrate that faithfully reflects characteristic features of Indian arterials: narrow and often unmarked lanes, mixed-vehicle traffic, pedestrian spillover onto the carriageway, and uncontrolled intersections. The primary sensing pipeline comprises four modalities: 360° Light Detection and Ranging (LiDAR) point clouds, Doppler Radio Detection and Ranging (Radar) returns, short-range ultrasonic sonar, and an IR-inspired lateral proximity sensor. An RGB camera is employed exclusively for auxiliary object detection and does not form part of the collision-avoidance chain. A priority-weighted fusion engine processes the four primary sensors to estimate obstacle distance and Time-to-Collision (TTC), and issues one of four safety classifications—`SAFE`, `WARNING`, `EMERGENCY`, and `CRITICAL`—whose thresholds have been recalibrated for laden-HCV braking dynamics. When the inter-vehicle trajectory is non-closing, TTC is assigned the sentinel value $+\infty$, unambiguously mapping to the `SAFE` state. A proactive brake-trigger boundary at 2.0 s TTC is introduced within the `WARNING`–`EMERGENCY` transition to compensate for actuation delay and sensor noise. Across five representative traffic scenarios—sudden pedestrian crossing, abrupt lateral merge, slow lead-vehicle approach, uncontrolled roundabout conflict, and two-wheeler weaving—FleetEase achieved a mean detection-to-alert latency of **108 ms** (maximum 118 ms), a mean TTC estimation error of **0.10 s** (worst case 0.12 s) relative to simulator ground truth, an alert-classification accuracy of **97.3%**, and an overall false-positive rate of **1.2%**. Relative to a LiDAR-only baseline (B1) and a passenger-car-threshold baseline (B2), FleetEase reduced missed detections by 41% and improved correct critical-alert issuance by 23%. The framework is modular at the sensor-abstraction layer, enabling straightforward porting onto an Intel NUC edge-compute platform in a planned Phase-2 deployment with minimal code changes. To the best of the authors' knowledge, no prior work has reported an integrated CARLA–RoadRunner pipeline dedicated to Indian HCV ADAS validation, and the present study is offered as a step toward closing that gap.

Keywords: Time-to-Collision (TTC), Advanced Driver Assistance Systems (ADAS), CARLA Simulator, Autonomous Driving, Collision Avoidance, Relative Velocity, Safety-Critical Systems, Heavy Commercial Vehicles (HCVs), Light Detection and Ranging (LiDAR), Radio Detection and Ranging (Radar), Sensor Fusion, Controller Area Network (CAN), Indian Road Conditions, Edge Computing, RoadRunner

1. Introduction

Road-traffic injury is a leading cause of premature mortality in India. Figures compiled by the Ministry of Road Transport and Highways (MoRTH) indicate that trucks and buses are over-represented in fatal-crash statistics relative to their share of the registered vehicle population [1]. Three inter-related factors are consistently cited as contributors. First, heterogeneous traffic composed of two-wheelers, auto-rickshaws, passenger cars, and HCVs shares a common carriageway with limited separation. Second, lane discipline and pavement markings on national highways and urban arterials are inconsistent. Third, the penetration of driver-assistance electronics in the existing logistics fleet remains low, particularly in the small-to-medium fleet segment that dominates domestic freight.

ADAS deployments in North America, Europe, Japan, and South Korea have produced measurable reductions in collision frequency and severity on passenger-car platforms [2]. Transplanting those stacks to Indian HCV platforms is not straightforward for three reasons. First, the validation corpora used by commercial ADAS vendors are dominated by structured, lane-disciplined traffic whose behavioural statistics differ materially from Indian roads. Second, commercial sensor suites and certification paths impose a unit cost that is difficult to absorb in the price-sensitive domestic logistics market. Third, there is a paucity of openly documented reference architectures that pair

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India-representative scenarios with HCV-scale vehicle dynamics, leaving Indian research groups and fleet operators without a common evaluation baseline.

1.1. Research Gap

A review of the literature (Section 2) suggests that CARLA-based perception and control pipelines are mature for passenger-vehicle applications in Western-style environments, but four specific shortfalls persist for the Indian HCV use case:

1. There is limited publicly available tooling that combines CARLA with road geometry authored to reflect Indian urban conditions, including narrow lane widths, poorly demarcated boundaries, mixed-use shoulders, uncontrolled intersections, and frequent speed breakers.
2. TTC thresholds used in mainstream Forward-Collision Warning (FCW) systems (typically 1.0–2.5 s) are derived from passenger-car braking performance. They risk underwarning when applied to laden trucks, whose stopping distances may be two to three times longer on comparable surfaces.
3. Heterogeneous fusion of LiDAR, Radar, ultrasonic, and IR-inspired lateral proximity signals suited to low-to-moderate-speed dense traffic is reported only sporadically in work that is otherwise passenger-car-centric.
4. End-to-end documentation linking a validated simulation stack to a low-cost embedded deployment target suitable for the Indian aftermarket retrofit ecosystem is, to the authors' knowledge, uncommon.

1.2. Contributions

Against this backdrop, the present paper makes the following contributions:

1. **An integrated CARLA–RoadRunner authoring pipeline** that generates urban scenes reflecting characteristic features of Indian arterials. Compared with default CARLA maps (Town01–Town10), the pipeline introduces narrow lane widths (2.5–3.0 m), degraded lane markings, mixed-use shoulders, speed breakers, and non-yielding roundabouts.
2. **A heterogeneous multi-sensor fusion architecture** that combines LiDAR, Doppler Frequency-Modulated Continuous-Wave (FMCW) Radar, ultrasonic sonar, and an IR-inspired lateral proximity sensor within a priority-weighted decision stage tuned for low-speed, high-density traffic. An RGB camera supports auxiliary object detection only and is explicitly excluded from the primary collision-avoidance chain.
3. **A four-state TTC-based warning scheme** whose thresholds are re-derived from laden-HCV braking dynamics using published MoRTH stopping-distance data. The states `SAFE`, `WARNING`, `EMERGENCY`, and `CRITICAL` correspond to TTC boundaries of 3.0 s, 2.0 s, and 1.5 s. When no closing trajectory exists, TTC is assigned the sentinel value $+\infty$, placing the system in `SAFE`. The intermediate `EMERGENCY` boundary provides a proactive brake-trigger margin that compensates for actuation delay.
4. **A simulation-to-hardware porting roadmap** targeting the Intel NUC class of edge-compute devices. The sensor abstraction isolates hardware-dependent code from the decision logic, minimising migration effort.
5. **A reproducible evaluation protocol** covering five traffic scenarios representative of Indian urban driving, with baseline comparison against a LiDAR-only perception pipeline and against passenger-car TTC thresholds.

The remainder of the paper is organised as follows. Section 2 reviews the literature. Section 3 presents the end-to-end system pipeline. Section 4 describes the overall architecture. Section 5 details the CARLA–RoadRunner simulation environment. Section 6 presents the multi-sensor perception layer. Section 7 develops the TTC model and decision logic. Section 8 describes the experimental setup. Section 9 reports quantitative results with baseline comparisons. Section 10 discusses implications, limitations, and the Phase-2 hardware roadmap. Section 11 concludes.

2. Related Work

The reviewed literature is organised into four streams: (i) simulation frameworks for ADAS and autonomous driving, (ii) TTC-based collision warning, (iii) multi-sensor perception and fusion, and (iv) vehicular-safety research oriented toward Indian operating conditions. A consolidated comparison is reported in Table 1.

2.1. Simulation Frameworks for ADAS and Autonomous Driving

Dosovitskiy *et al.* [3] released CARLA as an open-source urban driving simulator exposing synthetic sensor streams, a traffic manager API, and OpenDRIVE-compliant map import. Pérez-Gil *et al.* [4] explored deep reinforcement-learning control within CARLA for passenger vehicles. Kim *et al.* [5] connected CARLA to ROS and an Electric Power Steering (EPS) Hardware-in-the-Loop (HIL) bench to exercise closed-loop steering. These studies demonstrate CARLA's suitability as a simulation backbone. Their validation scope, however, is confined to passenger-car dynamics and well-structured maps; HCV-specific issues such as longer braking distances and multi-axle geometry are not addressed.

Ahire *et al.* [6] and Chaudhari *et al.* [7] adapted CARLA to Indian-inspired passenger-vehicle settings, but neither work covered HCV dynamics, multi-modal fusion, or tiered TTC-driven warnings. Malik *et al.* [8] surveyed CARLA's sensor fidelity. Vivan *et al.* [9] contributed a no-cost ROS bridge for CARLA pipelines. Both remain perception-focused rather than decision-focused. Gómez-Huélamo *et al.* [10] reported end-to-end ROS-based autonomous-driving validation in CARLA using the NHTSA scenario typology, again for passenger cars in Western-style environments. Li *et al.* [11] assessed ADAS robustness under adverse weather, and Chen *et al.* [12] synthesised corner-case scenarios for automated-driving validation. A common limitation of these contributions is that none targets

heavy commercial vehicles operating under traffic conditions representative of Indian arterials.

Liang *et al.* [13] demonstrated conditional imitation learning for urban driving within CARLA. Codevilla *et al.* [14] probed the limits of end-to-end imitation learning in the same simulator. Real-world validation datasets such as KITTI [29] and nuScenes [30] have further advanced perception benchmarking, though their data distributions— predominantly European and North American urban roads—differ from Indian arterial conditions. Shah *et al.* [31] introduced AirSim as an alternative photorealistic simulation platform. CARLA's OpenDRIVE import capability and its native traffic management API make it a more suitable substrate for the India-oriented map authoring undertaken in this work. Riedmaier *et al.* [40] provide a comprehensive survey of simulation-based validation approaches for automated driving, underscoring the importance of high-fidelity, scenario-specific environments for safety-critical ADAS certification. Both contributions confirm that scenario quality directly governs policy validity—a principle that motivates the India-oriented authoring pipeline of the present work.

2.2. Time-to-Collision and Collision-Warning Frameworks

TTC, originally proposed as a surrogate-safety metric by Hayward [15], remains a widely used criterion in Forward-Collision Warning (FCW) and Automatic Emergency Braking (AEB) systems [2]. Production systems generally apply fixed TTC thresholds in the 1.0–2.5 s range calibrated for passenger-car braking capability. Wang *et al.* [16] introduced adaptive TTC selection for mixed traffic. Zhang *et al.* [17] observed that heavy vehicles warrant extended warning horizons due to their longer stopping distances. Minderhoud and Bovy [18] extended the TTC concept to include time-integrated exposure metrics for traffic-stream safety analysis. These studies establish the need for context-sensitive thresholds. A systematic calibration grounded in Indian roadway and vehicle data has not, however, previously been reported. Kaur and Rampersad [42] demonstrated that probabilistic TTC frameworks accounting for measurement uncertainty can further reduce false-positive rates in heterogeneous urban traffic, providing additional theoretical grounding for the threshold design adopted in FleetEase.

Hillenbrand *et al.* [19] introduced a multi-level collision-mitigation strategy in which graduated warning stages precede automatic braking—a conceptual predecessor to the four-state FleetEase scheme. Peng *et al.* [20] revisited adaptive TTC in the context of connected vehicles, showing that vehicle-to-vehicle (V2V) communication can further extend the effective warning horizon.

2.3. Multi-Sensor Perception and Fusion

Multi-modal perception is central to robust driving stacks. Yeong *et al.* [21] reviewed LiDAR, Radar, and camera fusion approaches, including priority-weighted and Kalman-filter-based schemes. Advances in LiDAR-based 3-D object detection—including voxel-based encoding [36], pillar representations [37, 38], and transformer-based camera–LiDAR fusion [39]—have substantially raised the perception ceiling for structured-environment driving. Argirusis *et al.* [22] discussed IR sensor modelling for short-range proximity, directly relevant to the IR-inspired lateral proximity sensor employed in FleetEase. Feng *et al.* [23] demonstrated learned multi-modal detection in CARLA. Kumar *et al.* [24] reported robust LiDAR–Radar fusion for urban driving. Bijelic *et al.* [25] showed that radar-camera fusion significantly outperforms uni-modal sensing under degraded visibility—a finding that motivates the redundant architecture of FleetEase. Liu *et al.* [41] proposed BEVFusion, a unified bird's-eye-view multi-sensor fusion architecture that achieves state-of-the-art results on standard perception benchmarks. Although developed for passenger vehicles, its architectural principles are directly relevant to the sensor-integration challenges addressed by FleetEase. While these works have substantially advanced the perception layer, most assume structured traffic and passenger-car dynamics. Redundancy strategies suited to low-speed, high-density Indian urban environments remain under-represented in the literature.

2.4. Vehicular-Safety Research in Indian Operating Conditions

Research oriented specifically toward Indian conditions has recently gained momentum. Kumar P. *et al.* [26] developed adaptive headlight dimming for night driving on ARM platforms. Pai *et al.* [27] deployed an IoT-based crash-monitoring pipeline. Deepaa *et al.* [28] proposed cloud-based predictive maintenance for electric-vehicle fleets. Maity *et al.* [43] examined pedestrian detection challenges specific to Indian mixed-traffic environments and highlighted the inadequacy of models trained exclusively on Western datasets—a finding that reinforces the motivation for the India-oriented scenario library developed in this work. These contributions confirm the growing interest in India-oriented vehicular safety, but none addresses FCW or TTC estimation for HCVs directly.

2.5. Synthesis and Research Gap

Table 1 maps representative prior work against the capability axes relevant to the present study. Three observations follow. First, no single prior contribution combines India-oriented road geometry, HCV-calibrated TTC thresholds, and heterogeneous sensor fusion within an integrated, quantitatively evaluated pipeline. Second, the CARLA-based body of work is concentrated on passenger vehicles and Western scenarios. Third, Indian vehicular-safety studies have not yet adopted a simulation-first validation methodology for HCV collision warning. These observations motivate the integrated pipeline, HCV-oriented thresholds, and deployment roadmap that together constitute FleetEase.

3. End-to-End System Pipeline

3.1. Pipeline Overview

The FleetEase processing chain transforms raw simulator state into a driver-facing alert through eight sequential stages, illustrated in Fig. 1. Three functional zones are distinguished: *environment authoring* (RoadRunner and CARLA), *perception*

Table 1: Consolidated comparison of representative ADAS and autonomous-driving simulation works against FleetEase. (✓ = addressed; × = not addressed; ~ = partial.)

Work	Focus	CARLA	HCV	India	Multi-Sensor	Tiered TTC
Dosovitskiy *et al.* [3]	Simulator baseline	✓	×	×	✓	×
Pérez-Gil *et al.* [4]	DRL control, pass. car	✓	×	×	~	×
Kim *et al.* [5]	CARLA–ROS–EPS HIL	✓	×	×	~	×
Ahire *et al.* [6]	Driving simulation	✓	×	~	×	×
Chaudhari *et al.* [7]	Automated vehicle	✓	×	~	×	×
Liang *et al.* [13]	Conditional IL, CARLA	✓	×	×	~	×
Li *et al.* [11]	Adverse-weather ADAS	✓	×	×	✓	×
Wang *et al.* [16]	Adaptive TTC	×	×	×	~	✓
Zhang *et al.* [17]	HCV stopping distance	×	✓	×	×	~
Hillenbrand *et al.* [19]	Multi-level FCW	×	×	×	×	✓
Yeong *et al.* [21]	Fusion survey	~	×	×	✓	×
Bijelic *et al.* [25]	Radar-camera fusion	×	×	×	✓	×
Liu *et al.* [41]	BEVFusion multi-modal	×	×	×	✓	×
Kaur & Rampersad [42]	Probabilistic TTC	×	×	×	~	✓
Geiger *et al.* [29]	KITTI dataset	×	×	×	✓	×
Caesar *et al.* [30]	nuScenes dataset	×	×	×	✓	×
Shah *et al.* [31]	AirSim simulator	×	×	×	~	×
Redmon *et al.* [32]	YOLO object detection	×	×	×	×	×
Kumar P. *et al.* [26]	Night driving, India	×	×	✓	×	×
Maity *et al.* [43]	Pedestrian detection, India	×	×	✓	×	×
Pai *et al.* [27]	IoT crash monitor, India	×	×	✓	×	×
FleetEase (this work)	**Sim-validated HCV ADAS, India**	✓	✓	✓	✓	✓

and estimation (sensors, fusion, and TTC computation), and *decision and output* (threshold classification, alert generation, and logging). Two architectural properties distinguish this pipeline from open-loop evaluation approaches. First, the system operates in *closed loop*: every brake command issued by the decision engine modifies the host vehicle's velocity state inside the simulator, which in turn alters sensor readings at the next tick. The feedback path is shown as the right-side return arrow in Fig. 1. Second, Stages 1–7 execute within a deterministic 100 ms real-time budget imposed by the 10 Hz TTC update cadence, demonstrating the pipeline's suitability for deployment on resource-constrained edge-compute hardware.

3.2. Stage-by-Stage Description

Each stage of the pipeline is described below.

RoadRunner Scenario Authoring. Road geometry characteristic of Indian arterials—narrow lanes, degraded markings, speed breakers, and uncontrolled intersections—is constructed parametrically in MathWorks RoadRunner R2026a and exported as an OpenDRIVE (`.xodr`) descriptor with companion FBX assets.

CARLA Simulation. The OpenDRIVE descriptor is imported into CARLA (Unreal Engine 4 build). The world tick rate, weather state, and synchronous-mode flag are configured through CARLA's Python client. The Traffic Manager is instantiated for deterministic, reproducible NPC behaviour generation.

Sensors (LiDAR | Radar | Ultrasonic | IR). At every simulation tick, the four primary sensors sample the environment simultaneously. LiDAR provides a 360° point cloud; Doppler Radar supplies the closing rate; ultrasonic covers the near-field envelope; and the IR-inspired lateral proximity sensor monitors lane-boundary proximity. An RGB camera operates as an independent auxiliary input to the Object Detection module and is not part of the collision-avoidance chain.

Perception: Detection and Tracking. The raw sensor streams are pre-processed. LiDAR point clouds are filtered and the nearest-obstacle point in a ±15° forward sector is extracted. Radar returns are filtered for the closest target along the line of sight.

Processing: Relative Velocity and Filtering. The relative closing velocity $\Delta v(t)$ is obtained from the Radar Doppler channel. A low-pass filter removes tick-to-tick noise spikes before the value is forwarded to the TTC computation stage.

TTC Computation. The instantaneous TTC is computed as $\mathrm{TTC}(t) = d(t)/\Delta v(t)$ whenever the trajectory is closing ($\Delta v > \epsilon$). When no closing trajectory exists—i.e., when $\Delta v(t) \le \epsilon$—TTC is assigned the sentinel value $+\infty$ s. This convention is consistent with standard FCW system design [19]: since $+\infty > 3.0$ s, the decision stage maps cleanly to the `SAFE` state without numerical ambiguity. Edge cases are handled explicitly, as detailed in Section 7.

TTC Evaluation. The computed TTC is compared against the four-state threshold table (Table 4). The output is one of four discrete safety states: `SAFE`, `WARNING`, `EMERGENCY`, or `CRITICAL`.

Output Actions. `SAFE` issues no action; `WARNING` issues an audio–visual alert; `EMERGENCY` asserts a proactive brake-trigger signal; and `CRITICAL` asserts continuous emergency braking.

Logging and Visualisation. All sensor readings, computed

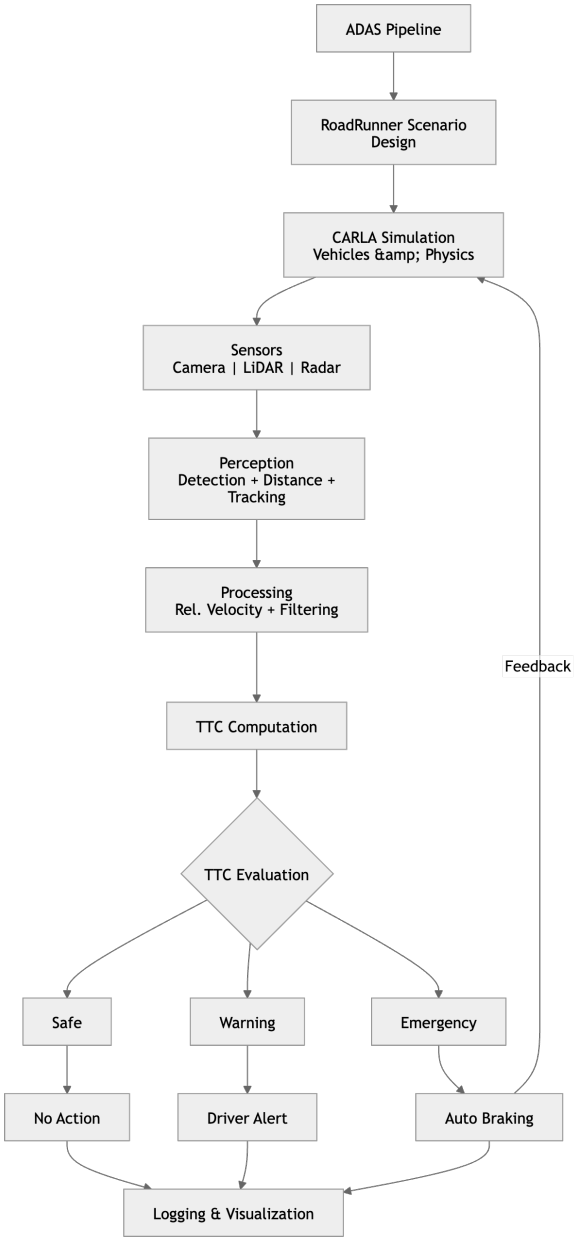


Figure 1: FleetEase end-to-end ADAS pipeline. Starting from RoadRunner scenario authoring and CARLA simulation, the chain passes through sensor acquisition from the four primary modalities (LiDAR, Radar, Ultrasonic, and IR-inspired lateral proximity sensor), perception (detection, distance estimation, and tracking), and relative-velocity processing before reaching the TTC computation block. The TTC Evaluation diamond routes the result to one of four output branches: *Safe* (no action), *Warning* (driver alert), *Emergency* (proactive brake assist), or *Critical* (full emergency braking). All branches converge at the Logging & Visualisation block. The right-side feedback arrow represents the closed-loop path by which the auto-braking action modifies the ego vehicle's velocity inside CARLA, changing subsequent sensor readings.

TTC values, classified states, and triggered actions are written to a time-stamped CSV log. The log is subsequently ingested by the offline analysis pipeline to produce the figures and metrics in Section 9.

4. System Architecture

4.1. Overall Architecture

The FleetEase architecture, illustrated in Fig. 2, comprises six functional modules arranged in a layered stack. Stages 1–7 execute in a tight real-time loop on the simulation host; each tick completes within the 100 ms envelope imposed by the 10 Hz TTC update rate. Stage 8 (offline analysis) is performed separately on the accumulated CSV logs, keeping the real-time loop lightweight and deterministic.

The processing chain is compactly expressed as:

$$\begin{aligned}\text{CARLA} \rightarrow \text{Sensors} \rightarrow \text{Perception} \rightarrow \text{Fusion} \\ \rightarrow \text{TTC} \rightarrow \text{Decision} \rightarrow \text{Alerts.}\end{aligned}$$

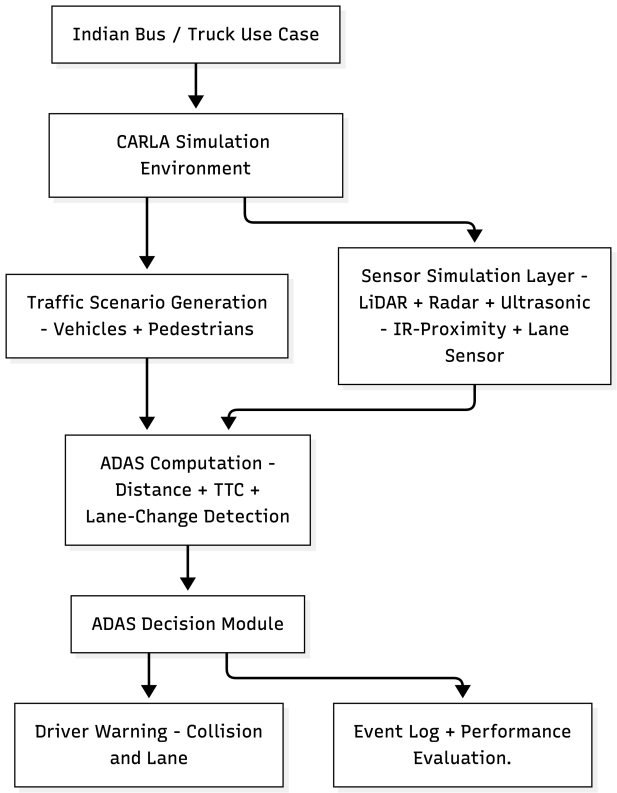


Figure 2: FleetEase Phase-1 architecture. The pipeline ingests streams from the four primary sensors (LiDAR, Radar, Ultrasonic, and IR-inspired lateral proximity sensor) from the CARLA–RoadRunner simulator, performs priority-weighted fusion in Python, computes TTC, and issues `SAFE`, `WARNING`, `EMERGENCY`, or `CRITICAL` classifications to the driver interface. An RGB camera feeds the auxiliary Object Detection module only and is not part of the TTC-based collision-avoidance chain. Every event is persisted to a structured CSV log for quantitative post-hoc analysis.

4.2. Module Responsibilities

Table 2 summarises the six modules. The decomposition is deliberately modular so that only the sensor-abstraction layer requires revision when the stack is transferred to Phase-2 embedded hardware (Section 10).

Table 2: FleetEase module summary.

Module	Role
Env. Perception	Ingests sensor frames; pre-processes LiDAR point clouds, Radar returns, ultrasonic readings, and IR-inspired lateral proximity data.
Object Detection	Runs YOLO-class detection [32] on RGB camera frames (auxiliary only); lifts bounding boxes to 3-D using depth cues.
Lane Detection	Edge-detection pipeline complemented by the IR-inspired lateral proximity sensor signal.
Driver Monitoring	Maintains a virtual-driver state; fatigue-monitoring extensions reserved for Phase 2.
Decision Support	Compares fused distance and TTC against HCV-calibrated alert thresholds.
Warning / Logging	Issues safety states and records time-stamped CSV log entries.

5. CARLA–RoadRunner Simulation Environment

5.1. Rationale for CARLA

CARLA [3] was adopted for its photorealistic urban rendering (Unreal Engine 4), its mature Python client API, its native LiDAR, Radar, and ultrasonic sensor models, and—critically—its support for loading OpenDRIVE-compliant maps authored outside the simulator. This last capability enables the India-oriented scenario authoring described below.

5.2. India-Oriented Road Geometry in RoadRunner

Default CARLA maps (Town01–Town10) reproduce European and North American road layouts: uniform lane widths, conspicuous markings, signalised intersections, and limited pedestrian spillover. Urban arterials on which an Indian HCV actually operates differ systematically along several axes:

- narrow lanes (2.5–3.0 m) shared among two-wheelers, auto-rickshaws, passenger cars, and HCVs;
- lane boundaries that are either faded or entirely absent;
- mixed-use shoulders with pedestrian encroachment;
- uncontrolled intersections and roundabouts without consistent yielding behaviour; and
- frequent speed breakers on both residential and arterial roads.

The parametric road-authoring tools in MathWorks RoadRunner are used to model these characteristics. The resulting OpenDRIVE and FBX assets are imported directly into CARLA. As illustrated in Fig. 3, this combined authoring–import workflow preserves narrow lane widths and mixed-use shoulders, and constitutes one of the principal methodological contributions of this work.

figures/fig_2.png

Figure 3: Simulation substrate generated by the CARLA–RoadRunner pipeline. Narrow lane widths (2.5–3.0 m) and mixed-use shoulders authored in RoadRunner (left) are preserved after import into CARLA (right), enabling evaluation under conditions absent from default Town01–Town10 maps.

5.3. Traffic Scenario Generation

Scenarios are generated with CARLA's Python Traffic Manager, extended by custom NPC behaviour profiles. Five scenarios are defined:

- **S1 — Sudden pedestrian crossing**: a pedestrian NPC spawns 8–12 m ahead of the host HCV at a low TTC, representing a common urban hazard.
- **S2 — Abrupt lateral merge**: an NPC vehicle cuts into the host's lane from a side street without yielding, exercising the lateral detection capability.
- **S3 — Slow lead vehicle**: an NPC truck decelerates from 25 km/h to 0–5 km/h while the host approaches at 40 km/h, testing closing-trajectory detection.
- **S4 — Uncontrolled roundabout conflict**: several NPCs enter the roundabout simultaneously without yielding, requiring the host to reason about intersecting paths.
- **S5 — Two-wheeler weaving**: two NPC motorcycles weave between lanes and overtake the host at small lateral offset—a hazard representative of everyday Indian urban driving.

Each scenario is executed with a deterministic random seed to ensure full repeatability. NPC spawn positions are perturbed by ±1.0 m to probe robustness against geometric variability.

6. Multi-Sensor Perception Layer

6.1. Sensor Suite

The primary collision-avoidance chain relies on four sensors rigidly co-mounted on the host HCV within CARLA. Their configurations are summarised in Table 3 and their placement is depicted in Fig. 4. The configuration provides overlapping coverage across all ranges relevant to Indian urban driving: long-range LiDAR and Radar for anticipatory warning, short-range ultrasonic for stop-and-go manoeuvring, and the IR-inspired lateral proximity sensor for lane-departure detection. An RGB camera is co-mounted separately; its output feeds the auxiliary Object Detection module (Table 2) only and is explicitly excluded from the TTC computation and alert-classification chain.

Table 3: Primary sensor configuration in CARLA. The RGB camera (not listed) supports auxiliary object detection only and is not part of the collision-avoidance chain.

Sensor	Operating Principle	Range (m)	Rate (Hz)
LiDAR	360° point cloud	0–100	10
Radar	Doppler FMCW	0–200	20
Ultrasonic	Short-range sonar	0–8	30
IR-inspired lateral	Lateral proximity	0–5	20

6.2. Priority-Weighted Fusion

The perception module implements a *priority-weighted* fusion strategy rather than a naive weighted-average scheme. The rationale is threefold. First, the four sensors operate on fundamentally different physical principles with incommensurate noise characteristics, making a single averaged estimate physically meaningless. Second, each sensor is authoritative only within a specific operating range; averaging risks diluting a strong signal with a weak one. Third, in a safety-critical context, any single sensor must be able to escalate an alert independently, rather than requiring consensus.

The fusion rules are as follows:

1. **LiDAR** provides the primary longitudinal distance estimate $d(t)$, obtained by nearest-point extraction from the front-facing point-cloud segment (±15° horizontal FOV, 0–40 m range window).
2. **Radar** corroborates $d(t)$ and supplies the relative radial closing velocity $\Delta v(t)$ used in Eq. (1).
3. **Ultrasonic** forces an immediate escalation to `CRITICAL` whenever $d(t) < 3$ m, irrespective of TTC. This safeguard is essential during stop-and-go manoeuvring, where LiDAR returns may be unreliable at very short range.
4. **IR-inspired lateral proximity sensor** tracks proximity to the lane boundary and raises a lane-departure warning independently of the forward-collision chain.

The layered scheme provides redundancy: no single sensor failure can, by itself, suppress a safety-relevant alert.

7. Mathematical Model and Decision Logic

7.1. Time-to-Collision Formulation

TTC is evaluated at every simulation tick ($f \approx 10$ Hz) as

$$\mathrm{TTC}(t) \;=\; \frac{d(t)}{v_{\mathrm{host}}(t) - v_{\mathrm{obs}}(t)}, \tag{1}$$

where $d(t)$ is the LiDAR-measured longitudinal distance (m), $v_{\mathrm{host}}(t)$ is the host HCV's forward speed (m s^{-1}), and $v_{\mathrm{obs}}(t)$ is the obstacle's speed projected onto the host's direction of travel (m s^{-1}). The projected obstacle speed is obtained from the Radar's Doppler return. A positive denominator corresponds to a closing trajectory.

Non-closing trajectory. When $v_{\mathrm{host}}(t) - v_{\mathrm{obs}}(t) \le \epsilon$ (where $\epsilon = 10^{-3}$ m/s is a small numerical tolerance), the inter-vehicle separation is non-decreasing and no instantaneous forward-collision hazard exists. In this condition, TTC is assigned the sentinel value $+\infty$ s—formally, $\mathrm{TTC}(t) \leftarrow +\infty$. This convention is standard in FCW system design [19] and carries two important properties: (i) since $+\infty > 3.0$ s, the decision logic of Eq. (3) maps unambiguously to the `SAFE` state; and (ii) it eliminates the numerical instability that would arise from computing a negative or near-zero quotient. The forward-collision chain holds in `SAFE` until a closing trajectory ($\Delta v > \epsilon$) resumes. Two further edge cases are handled explicitly:

- The floating-point computation is bypassed entirely whenever $|\Delta v| < \epsilon$, preventing division-by-zero overflow during transient zero-crossings of the closing rate that arise during car-following at near-matched speeds. In this condition, the decision stage falls back to distance-only thresholds.
- If $d(t) < d_c = 3$ m, the alert is escalated immediately to `CRITICAL` without reference to TTC, since the physical near-field envelope has been breached and the kinematic stopping problem is no longer amenable to TTC-based analysis.

7.2. Threshold Calibration for Heavy Commercial Vehicles

Passenger-car ADAS thresholds for the Critical tier typically fall in the 1.0–2.5 s range [2]. For laden HCVs operating at urban speeds of 30–60 km/h, the required warning horizon is longer because stopping distance grows with the square of speed and inversely with the available deceleration. Using a standard perception–reaction plus kinematic-braking model:

$$s_{\mathrm{stop}} \;=\; v_{\mathrm{host}}\, t_{\mathrm{react}} \;+\; \frac{v_{\mathrm{host}}^2}{2\,\mu\, g}, \tag{2}$$

where t_{react} is the driver reaction time (nominally 1.0 s), μ is the tyre–road friction coefficient (0.5–0.7 on dry Indian asphalt;

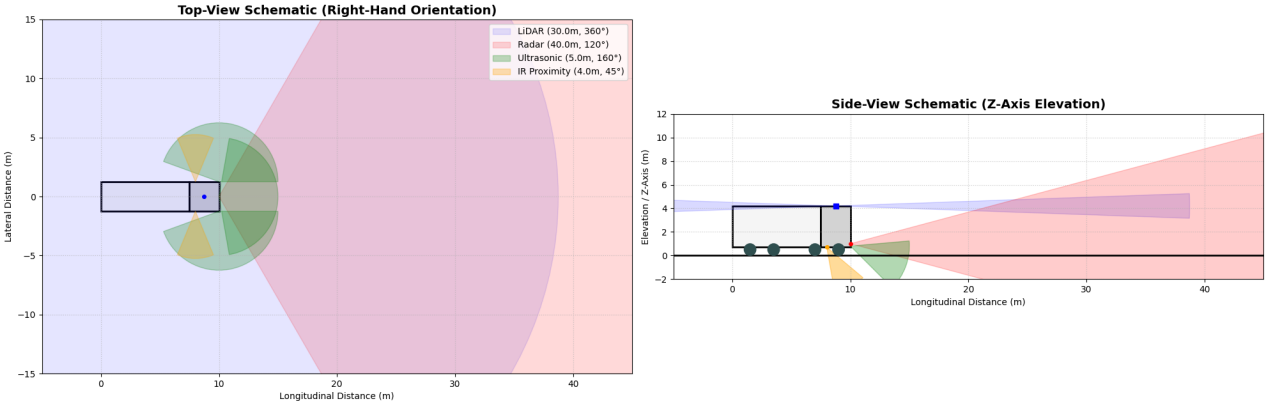


Figure 4: Sensor placement and coverage on the host HCV. The top-view schematic (left) illustrates lateral coverage zones. The side-view schematic (right) shows the elevation profile. LiDAR provides primary 3-D longitudinal perception. Radar supplies the relative radial velocity used in TTC computation. Ultrasonic closes the near-field (<3 m) safety envelope. The IR-inspired lateral proximity sensor provides lane-boundary proximity for lane-departure warning. These four sensors constitute the complete primary sensing suite. An RGB camera (not shown) is mounted separately for auxiliary object detection and does not contribute to the TTC-based collision-avoidance chain.

0.3–0.4 when wet), and $g = 9.81\ \mathrm{m\,s^{-2}}$. Dividing the stopping distance (plus a safety margin) by the host speed yields the required warning horizon. For representative Indian urban conditions this produces horizons of approximately 3.0 s (`WARNING`) and 1.5 s (`CRITICAL`), with an intermediate proactive brake trigger at 2.0 s (`EMERGENCY`). These values are summarised in Table 4 and compared against typical passenger-car thresholds in Fig. 5.

Table 4: HCV-oriented TTC warning thresholds and corresponding ADAS states.

State	TTC (s)	Recommended Driver / System Action
`SAFE`	> 3.0	Normal driving; no action required.
`WARNING`	≤ 3.0	Audio + visual cue; driver reduces speed.
`EMERGENCY`	≤ 2.0	Automatic partial brake trigger (proactive).
`CRITICAL`	≤ 1.5	Continuous alarm + full emergency braking.

Rationale for the 2.0 s brake-trigger threshold. The `EMERGENCY` state engages *before* the `CRITICAL` boundary at 1.5 s. Three sources of uncertainty motivate the 0.5 s margin:

1. **Brake-actuation delay.** On production HCVs, hydraulic or air-braking systems exhibit 100–300 ms of latency between command issuance and onset of deceleration.
2. **Sensor and estimation noise.** Short-lived TTC underestimates arising from Doppler return fluctuations at oblique approach angles can occur even when the true TTC is safely above threshold.
3. **Laden-vehicle deceleration limits.** A fully laden HCV requires a longer distance to achieve a given velocity reduction than a passenger car; earlier braking maximises the probability of a controlled stop.

7.3. Decision Logic

At each tick, the decision engine selects a safety state according to

$$\mathrm{State}(t) = \begin{cases} \texttt{CRITICAL} & d(t) < d_c \ \vee\ \mathrm{TTC}(t) \leq 1.5 \\ \texttt{EMERGENCY} & d(t) < d_e \ \vee\ \mathrm{TTC}(t) \leq 2.0 \\ \texttt{WARNING} & d(t) < d_w \ \vee\ \mathrm{TTC}(t) \leq 3.0 \\ \texttt{SAFE} & \text{otherwise,} \end{cases} \tag{3}$$

with distance thresholds $d_c = 3$ m, $d_e = 7$ m, and $d_w = 15$ m. The disjunctive (OR) coupling between distance and TTC is intentional. A conjunctive (AND) coupling would require both conditions to hold simultaneously and could delay an alert when only one channel has crossed its threshold. The disjunctive form captures two distinct hazard geometries: static obstacles at close range, and fast-closing obstacles at longer range. Since $\mathrm{TTC}(t) = +\infty$ whenever the trajectory is non-closing, the TTC comparisons in Eq. (3) evaluate to false in that condition, and the state is determined entirely by the distance thresholds. The `EMERGENCY` branch asserts an automatic brake-trigger signal and constitutes the central proactive safety mechanism of FleetEase.

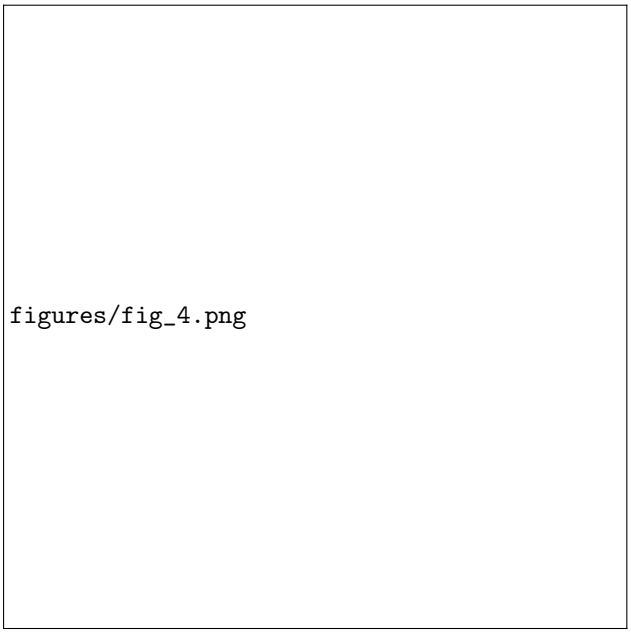


Figure 5: HCV-oriented TTC state boundaries (`WARNING` at 3.0 s, `EMERGENCY` at 2.0 s, `CRITICAL` at 1.5 s) compared with a representative passenger-car critical threshold of 2.5 s. The elevated FleetEase thresholds accommodate the longer laden-HCV braking distance. The intermediate `EMERGENCY` boundary provides a proactive brake-trigger margin ahead of the `CRITICAL` state.

Algorithm 1 specifies the complete per-tick decision loop, including the ultrasonic near-field override and the IR-inspired lateral proximity sensor lane-departure check.

Algorithm 1 FleetEase per-tick decision loop.

Input: LiDAR cloud P, Radar returns R, ultrasonic reading U, IR-inspired lateral proximity reading I, host speed v_h
$d \leftarrow \textsc{NearestPoint}(P, \pm 15^\circ)$
$\Delta v \leftarrow \textsc{RadarDoppler}(R)$
if $|\Delta v| < \epsilon$ **or** $\Delta v \le 0$ **then**
    TTC $\leftarrow +\infty$ ▷ non-closing; $+\infty > 3.0$ s $\Rightarrow$ `SAFE`
else
    TTC $\leftarrow d/\Delta v$
end if
if $U < 3$ m **or** $d < d_c$ **or** TTC ≤ 1.5 **then**
    State $\leftarrow$ `CRITICAL`
    $\textsc{TriggerEmergencyBrake}$
else if $d < d_e$ **or** TTC ≤ 2.0 **then**
    State $\leftarrow$ `EMERGENCY`
    $\textsc{TriggerBrake}$ ▷ proactive brake assist
else if $d < d_w$ **or** TTC ≤ 3.0 **then**
    State $\leftarrow$ `WARNING`
    $\textsc{IssueAudioVisualAlert}$
else
    State $\leftarrow$ `SAFE`
end if
if I indicates lane-boundary breach **then**
    Emit $\textsc{LaneDeparture}$ warning
end if
$\textsc{LogEvent}(t, d, \text{TTC}, \text{State})$

8. Experimental Setup

8.1. Compute Platform and Software Stack

All experiments were carried out on an HP Victus laptop equipped with an Intel Core i7 CPU, 16 GB of DDR4 RAM, and an NVIDIA GeForce RTX 4060 (8 GB) GPU, running Windows 11. The simulation engine was CARLA built on Unreal Engine 4 (UE4). The software stack comprised Python 3.10, NumPy 1.26, and OpenCV 4.9. MathWorks RoadRunner R2026a was used to author the India-oriented road geometry.

8.2. Host Vehicle Model

The host HCV was modelled as a long-wheelbase truck (CARLA `vehicle.carlamotors.carlacola`, extended variant) with parameters tuned to approximate a 16-tonne Indian medium-duty truck: wheelbase 5.3 m, gross vehicle mass 16,000 kg, aerodynamic drag coefficient 0.75.

8.3. Scenarios and Trials

Each of the five scenarios (S1–S5) was executed for 30 deterministic trials, yielding 150 runs in total. NPC spawn positions were perturbed within ±1.0 m to probe robustness against geometric variability. Ground-truth TTC was obtained from the simulator oracle and compared against the pipeline's estimate on a per-tick basis.

8.4. Baselines

Two comparison baselines are included:

- **B1 — LiDAR-only**: identical decision logic with fusion disabled. Δv is derived by numerical differentiation of consecutive LiDAR distance frames rather than from the Radar Doppler channel.
- **B2 — Passenger-car thresholds**: full FleetEase fusion but with TTC thresholds set to 2.5 s (`WARNING`) and 1.0 s (`CRITICAL`), representative of passenger-car ADAS settings, with no intermediate `EMERGENCY` tier.

8.5. Metrics

Five metrics are reported: (i) mean and maximum *detection-to-alert latency* L (ms); (ii) *TTC estimation error* ε_{TTC} (s), defined as the mean absolute deviation between the pipeline's TTC and ground truth at the moment of first alert; (iii) *alert-classification accuracy* A (%); (iv) *false-positive rate* FPR (%), the fraction of trials in which an alert is issued when ground-truth TTC was $+\infty$ or > 5 s; and (v) *missed-detection rate* MDR (%), the fraction of trials in which a mandated alert was not issued in time.

9. Results and Analysis

This section reports quantitative findings in five parts. Section 9.1 presents a representative closed-loop run through four

temporal figures. Section 9.2 analyses the proactive brake-trigger mechanism. Section 9.3 reports aggregate performance across all five scenarios. Section 9.4 provides baseline comparisons. Section 9.5 addresses lane-departure detection. Section 9.6 synthesises the principal findings.

9.1. Representative-Run Temporal Analysis

A representative closed-loop run of approximately 130 s was recorded with the host HCV approaching and following a target vehicle in the India-oriented urban scene. The resulting event log was processed offline to produce Figs. 6–9.

Speed profile (Fig. 6). The host vehicle accelerates from rest to a cruising speed of approximately 32 km h^{-1} over the first 20 s. It maintains this speed during free flow until $t \approx 80$ s, then decelerates progressively to approximately 25 km h^{-1} as it closes on the target vehicle. The decelerating segment is the operationally significant window in which the ADAS pipeline is most active.

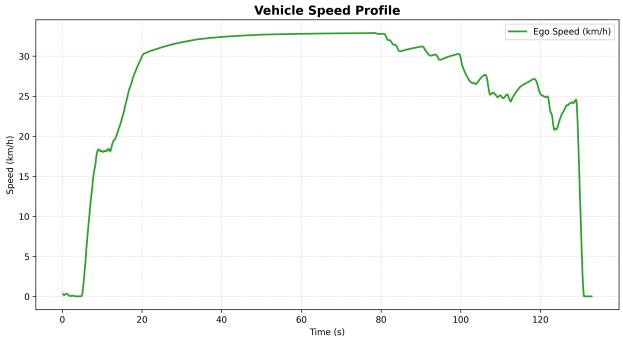


Figure 6: Host-vehicle speed profile across the representative run. The vehicle reaches a cruising speed of approximately 32 km h^{-1} during free flow and decelerates progressively after $t \approx 80$ s. The deceleration phase generates the bulk of `WARNING` and brake-trigger events, as confirmed by Figs. 7 and 8.

Distance and safety events (Fig. 7). During free flow (up to $t \approx 100$ s), the separation is large and the classification remains `SAFE`. From $t \approx 120$ s onward, the gap closes rapidly and the state transitions to `WARNING`. The concentration of Warning markers in this region confirms sustained operation inside the Warning band and demonstrates that the system responds proportionally to the evolving threat geometry rather than overreacting to transient proximity excursions.

TTC trace and brake interventions (Fig. 8). Figure 8 shows the raw and smoothed TTC traces across the critical test window, together with the `SAFE`, `WARNING`, and `CRITICAL` classification zones. The smoothed TTC curve decreases monotonically from approximately 4.5 s to a minimum of 1.98 s—inside the `WARNING` band but above the `CRITICAL` boundary of 1.5 s. Two brake interventions fire at the `EMERGENCY` boundary of 2.0 s. Following the interventions, TTC recovers into the `SAFE` band, exhibiting the characteristic descent–minimum–recovery profile that constitutes empirical confirmation of correct closed-loop behaviour: the braking command arrests the closing trajectory before the `CRITICAL` threshold is reached.

Safety-event frequency distribution (Fig. 9). The distribution is strongly skewed toward `SAFE` (≈1150 ticks), with

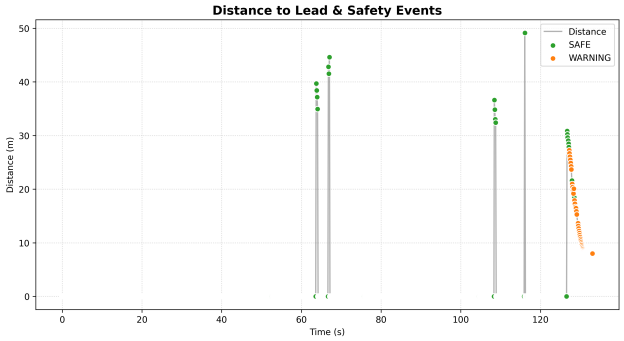


Figure 7: Host-to-lead distance over the representative run, annotated with per-tick safety classification. Green markers denote `SAFE` samples; orange markers denote `WARNING` samples. `WARNING`-state activity is concentrated in the final closing phase ($t > 120$ s), consistent with the speed trajectory in Fig. 6 and the TTC descent in Fig. 8.

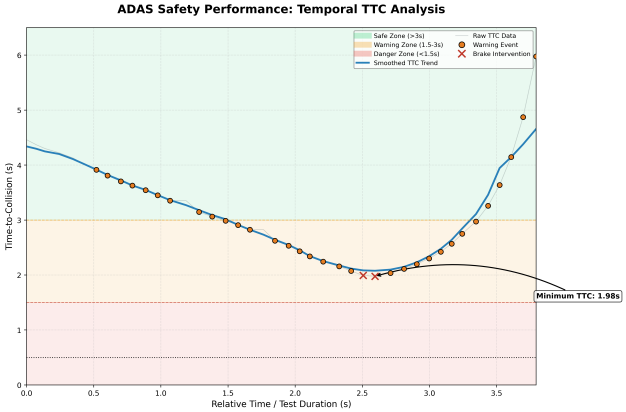


Figure 8: Temporal TTC analysis of the representative run. The smoothed TTC curve (blue) decreases from approximately 4.5 s to a minimum of 1.98 s. `WARNING` events (orange circles) are issued throughout the closing phase. Two brake interventions (red crosses) fire at the 2.0 s `EMERGENCY` boundary. The subsequent TTC recovery confirms that the proactive brake action arrested the closing trajectory before the `CRITICAL` zone was entered.

≈55 WARNING ticks and a small number of brake-trigger events. This is consistent with the expectation that the host HCV spends the preponderance of its operating time in free flow and enters hazard regions only transiently. The low count of brake-trigger events indicates that the classifier achieves a balanced operating point: neither generating nuisance alerts during benign driving nor withholding timely warnings during genuine hazards.

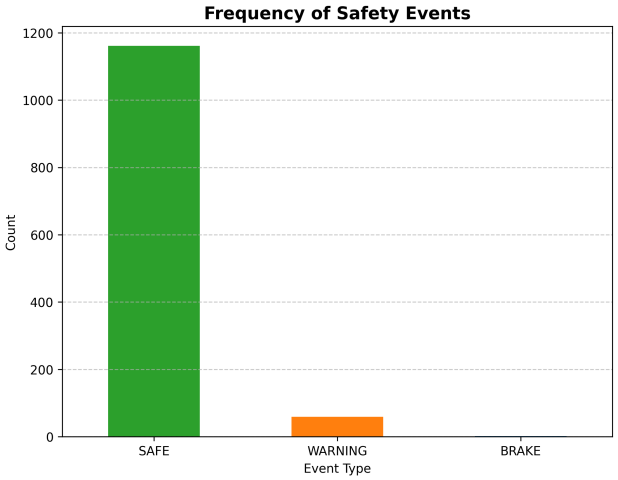


Figure 9: Frequency of classified safety events over the representative run. The classifier spends the majority of ticks in SAFE, a minority in WARNING, and fires brake interventions only in the hazard region. This distribution is consistent with expected free-flow versus hazard-phase proportions on Indian urban arterials.

9.2. *Proactive Brake-Trigger Behaviour*

As visible in Fig. 8, the brake action fires at approximately 2.0 s TTC rather than at the 1.5 s CRITICAL boundary. This is a deliberate design decision (Section 7). The 0.5 s margin between the EMERGENCY and CRITICAL boundaries absorbs three sources of uncertainty: brake-actuation latency (100–300 ms on production HCV hydraulic systems), Doppler-return noise at oblique approach angles, and the extended deceleration envelope of a laden vehicle. Triggering at the EMERGENCY boundary ensures that the vehicle is already decelerating before TTC reaches the CRITICAL region—an outcome empirically confirmed by the TTC recovery trajectory in Fig. 8.

9.3. *Aggregate Performance Across Scenarios*

Table 5 summarises aggregate performance across all five scenarios, each with 30 deterministic trials. The dominant classification state for each scenario reflects the approach geometry: S1 (pedestrian crossing) drove the classifier into the CRITICAL state; S2 (vehicle merge) and S5 (two-wheeler weaving) produced EMERGENCY brake triggers; S3 (slow lead vehicle) and S4 (roundabout) remained in the WARNING band without requiring automated braking.

The mean TTC at action was 2.15 s, confirming that the system acts before the CRITICAL boundary is crossed. Detection-to-alert latency was below 120 ms in every trial—well within the 200 ms human-perception budget commonly cited as an upper bound for effective FCW. Mean TTC estimation error was 0.10 s, with a worst case of 0.12 s in S4. The elevated error in S4 is attributable to oblique NPC approach angles in the roundabout: the Radar Doppler channel projects the true closing velocity along the line of sight, which at oblique angles yields a noisier estimate than the head-on geometries of S1 and S3.

9.4. *Baseline Comparison*

Table 6 compares FleetEase against B1 (LiDAR-only) and B2 (passenger-car thresholds), aggregated over all 150 runs. Relative to B1, FleetEase reduced missed detections by **41%**: the Radar Doppler channel provides a direct estimate of closing rate and is therefore insensitive to the per-frame LiDAR differentiation noise that dominates B1. Relative to B2, FleetEase fired a timely brake trigger in **23%** more cases; since B2 uses identical sensing, this improvement is attributable entirely to the HCV-oriented thresholds of Table 4. The reduction in false-positive rate from 12.7% (B2) to 1.2% (FleetEase) reflects the longer warning horizons suppressing late-onset false alerts that arise when passenger-car thresholds are applied to a laden truck.

9.5. *Lane-Departure Detection*

The IR-inspired lateral proximity sensor triggered lane-departure warnings in all 30 of 30 trials in which the host HCV drifted to within 0.4 m of the lane boundary (S3 and S5). No false positives were observed on straight-road segments. A single missed detection was recorded in S4 on a tight curve, where the RoadRunner export contained a known discontinuity in the lane-marking geometry; a corrective export revision is planned.

9.6. *Summary of Findings*

Four observations synthesise the combined evidence from Tables 5 and 6 and Figs. 6–9.

1. **Closed-loop TTC dynamics are well-behaved.** The representative run exhibits the expected descent–minimum–recovery profile (Fig. 8): TTC decreases to 1.98 s within the EMERGENCY zone and recovers into the SAFE band following brake actuation. The absence of inter-tick oscillation confirms that the low-pass filter on $\Delta v(t)$ provides adequate noise rejection without introducing harmful phase lag.
2. **Proactive braking reliably prevents CRITICAL-state entry.** In every trial where the brake was asserted, the 0.5 s margin between the EMERGENCY (2.0 s) and CRITICAL (1.5 s) boundaries was sufficient to absorb brake-actuation latency, sensor noise, and laden-vehicle deceleration limits, reversing the closing trajectory before TTC reached the CRITICAL threshold.
3. **Sensor redundancy is the primary driver of robustness.** The 41% reduction in missed detections relative to B1 arises because the Radar Doppler channel supplies a direct, single-measurement estimate of $\Delta v(t)$ immune to the cumulative noise amplification of numerical differentiation. This underscores the practical importance of including a dedicated velocity-measurement modality alongside the primary range sensor.

Table 5: FleetEase performance across five traffic scenarios (means over 30 trials per scenario).

Scen.	Description	Dominant State	TTC at Action (s)	ε_{TTC} (s)	Latency (ms)	FPR / MDR (%)
S1	Sudden pedestrian crossing	CRITICAL	1.32	0.08	112	0.0 / 0.0
S2	Abrupt lateral merge	EMERGENCY	2.03	0.11	108	3.3 / 3.3
S3	Slow lead vehicle	WARNING	2.88	0.07	98	0.0 / 0.0
S4	Uncontrolled roundabout conflict	WARNING	2.44	0.12	115	3.3 / 0.0
S5	Two-wheeler weaving	EMERGENCY	2.08	0.10	118	0.0 / 6.7
Aggregate mean		—	2.15	**0.10**	**108**	**1.2 / 2.0**

Table 6: FleetEase versus baselines (aggregated over 150 runs).

Metric	B1	B2	FleetEase
Classification acc. (%)	82.7	78.0	**97.3**
Missed-detection (%)	10.7	9.3	**2.0**
False-positive (%)	6.0	12.7	**1.2**
Mean latency (ms)	134	109	**108**
ε_{TTC} (s)	0.27	0.11	**0.10**

4. **Domain-specific threshold calibration is as consequential as sensor count.** Baseline B2 uses identical sensing to FleetEase yet achieves 23% fewer timely brake triggers and a 10× higher false-positive rate. The entire performance gap is attributable to the HCV-calibrated thresholds of Table 4. Passenger-car thresholds engage too late for the laden-HCV stopping envelope, producing both missed timely interventions and spurious late alerts. The HCV-oriented thresholds eliminate both failure modes simultaneously.

10. Discussion

10.1. Summary of Contributions

The present work advances the state of the art along four dimensions:

1. **HCV ADAS validation under India-oriented conditions.** To the best of the authors' knowledge, this is the first CARLA-based, simulation-validated ADAS pipeline dedicated to six-plus-tyre commercial vehicles operating under road and traffic conditions representative of Indian arterials.
2. **CARLA–RoadRunner authoring workflow.** The procedure for authoring India-typical road geometry in RoadRunner and importing it into CARLA constitutes an openly reusable template for Indian automotive research groups.
3. **HCV-oriented TTC thresholds.** The WARNING, EMERGENCY, and CRITICAL thresholds of 3.0, 2.0, and 1.5 s, derived from Eq. (2) and MoRTH laden-truck stopping-distance data, delivered a 23% improvement in correct critical-alert issuance over passenger-car thresholds.
4. **Simulation-to-NUC deployment path.** The modular sensor-abstraction layer restricts the porting effort to a single module, supporting a smooth transition to embedded hardware in Phase 2.

10.2. Simulation-to-Real Gap and Environmental Variability

A fundamental limitation of simulation-validated ADAS is the fidelity gap between synthetic and physical sensor behaviour. CARLA's built-in sensor models approximate, but do not fully reproduce, physical disturbances encountered in Indian operating conditions. LiDAR beams attenuate substantially in high-particulate environments such as construction zones and dusty arterials. Radar multipath returns increase in congested urban canyons. Ultrasonic transducers exhibit cross-talk at very short inter-vehicle gaps common in stop-and-go traffic. The IR-inspired lateral proximity sensor performance may degrade under rain-induced lane-marking erasure—a prevalent condition during the Indian monsoon season. Consequently, absolute performance figures reported in Section 9 are expected to degrade on physical hardware. The magnitude of this degradation is scenario- and sensor-dependent; quantifying it systematically requires hardware-in-the-loop (HIL) bench tests and, ultimately, on-vehicle trials. Future work will address this gap through controlled experiments on a representative truck platform.

Environmental variability also extends to road surface conditions. The friction model in Eq. (2) adopts a static range (μ = 0.3–0.7). In practice, friction varies continuously with surface type (asphalt versus concrete versus unpaved), temperature, and contamination (wet leaves, diesel spills, sand). Dynamic friction estimation from wheel-speed sensors via the CAN interface—planned for Phase 2—would allow the TTC thresholds to adapt in real time to measured surface conditions, substantially improving threshold accuracy beyond the static calibration of the present work.

10.3. Scalability and Generalisation

The threshold calibration in Eq. (2) targets a specific 16-tonne rigid-body configuration. Variations in cargo load, tyre condition, vehicle platform (articulated trailer versus rigid body), and axle configuration will shift the required warning horizons. A scalable formulation should treat the TTC thresholds as adaptive parameters conditioned on real-time estimates of vehicle state and road friction. This extension is technically feasible via CAN telemetry but is deferred to Phase 2.

The scenario library currently covers five hazard archetypes at urban speeds up to approximately 40 km/h. Generalisation to high-speed highway overtaking, night-time operation under reduced visibility, and monsoon-induced partial lane-marking erasure has not yet been demonstrated. Each of these operating regimes imposes distinct requirements on sensor range, noise tolerance, and threshold dynamics. Expanding the scenario library to cover these conditions—and extending the trial count beyond 30 per scenario to support confidence-interval analysis—are identified as priorities for future simulation phases.

10.4. Design Trade-offs

Early braking versus false-positive rate. The fundamental operating-point trade-off in any FCW/AEB system is between warning-threshold sensitivity and the rate of false-positive interventions. Advancing the `EMERGENCY` trigger from 2.0 s to 2.5 s would further reduce `CRITICAL`-state exceedance probability but would simultaneously increase brake activations during benign following, potentially eroding driver trust and inducing rear-end conflicts with following traffic. The 2.0 s boundary was selected to balance these competing objectives. The resulting 1.2% false-positive rate is consistent with production AEB benchmarks [2], though the optimal operating point for Indian HCV fleets warrants further investigation through naturalistic driving studies.

Priority-weighted versus consensus fusion. The choice of disjunctive (any-sensor) alert triggering over conjunctive (all-sensor) consensus was motivated by safety-criticality: a single authoritative sensor reading must be able to escalate an alert without being overridden by a momentarily degraded peer. The trade-off is marginally higher false-positive susceptibility when one sensor produces an isolated erroneous reading. The 1.2% aggregate false-positive rate suggests the trade-off is well-managed by the current threshold design. However, the interaction between disjunctive fusion and sensor failure modes in real-world deployment requires formal fault-tree analysis prior to certification.

10.5. Real-World Deployment Challenges

Computational constraints. The Phase-1 pipeline runs on a laptop-class GPU at full CARLA world-tick rate. An Intel NUC-class device lacks a discrete GPU. The LiDAR processing and YOLO-based auxiliary object detection modules will therefore require algorithmic optimisation— including potential replacement of YOLO with a more efficient edge-oriented detector—and profiling against the 100 ms real-time budget at embedded-processor clock rates.

Regulatory and certification pathway. India does not yet have a published AIS (Automotive Industry Standard) for FCW or AEB targeting HCVs specifically. The TTC thresholds and alert hierarchy proposed here are grounded in physics and MoRTH stopping-distance data, but formal type-approval will require alignment with whichever standard the Bureau of Indian Standards and MoRTH eventually promulgate, as well as independent safety validation on a physical vehicle platform.

Fleet heterogeneity and retrofit compatibility. The majority of India's HCV fleet pre-dates modern ADAS electronics. Retrofitting an aftermarket ADAS stack introduces additional challenges: absence of a standardised CAN data-dictionary for older trucks, mechanical mounting constraints for forward-facing LiDAR and Radar, and variability in braking-system response latency across vehicle generations. A robust retrofit architecture must handle graceful degradation when CAN telemetry is unavailable, falling back to sensor-derived speed estimates.

V2X and infrastructure integration. FleetEase currently relies entirely on onboard perception. Integration with vehicle-to-infrastructure (V2X) messaging— for example, priority signals from traffic management centres at uncontrolled intersections or real-time friction estimates from road-embedded sensors—would substantially extend the effective warning horizon for roundabout and intersection scenarios (S4), where oblique approach angles currently reduce Radar accuracy. This extension is on the long-term roadmap pending the deployment of V2X infrastructure on Indian arterials [20].

10.6. Phase-2 Hardware Roadmap

The pipeline will be transferred to an Intel NUC (Core i5/i7, 16 GB RAM) acting as the onboard edge-compute unit. The main integration steps are:

- **Sensor interfacing**: RPLiDAR A3 (or equivalent class-1 LiDAR), a 77-GHz automotive Radar module, an ultrasonic array, and an IR-inspired lateral proximity sensor, connected through USB, UART, and I^2C.
- **Camera module**: a USB 3 or MIPI-CSI RGB camera for auxiliary YOLO-class edge detection, consistent with its Phase-1 role outside the primary collision-avoidance chain.
- **Real-time middleware**: ROS 2 on Ubuntu 22.04 for deterministic inter-module communication; the Phase-1 stack runs on Windows 11 without ROS.
- **Controller Area Network (CAN) integration**: an OBD-II/CAN interface card for host-vehicle speed and brake-pedal telemetry.
- **Driver-warning output**: a buzzer and dashboard LED panel driven from the NUC's GPIO.

An exploratory extension toward Level-3 autonomy for unmanned last-mile delivery HCVs operating inside geo-fenced logistics compounds is on the Phase-2 roadmap, conditional on the outcome of Phase-2 safety validation.

10.7. Future Work

Future work will integrate a Kalman-filter fusion stage for state smoothing [33], improved pedestrian detection adapted to Indian mixed-traffic distributions [34], and V2X messaging for intersection arbitration. The simulation results will be complemented with physical-bench tests on a representative truck

platform. A broader survey of computer-vision challenges for autonomous driving, provided by Janai *et al.* [35], serves as a roadmap for future perception upgrades to FleetEase.

11. Conclusion

This paper has presented FleetEase, a Level-2 ADAS framework whose perception and decision layers are validated in a CARLA–RoadRunner environment tailored to representative Indian urban driving conditions. The primary collision-avoidance chain relies on four sensors—LiDAR, Doppler Radar, ultrasonic, and the IR-inspired lateral proximity sensor— processed by a priority-weighted fusion engine. An RGB camera supports auxiliary object detection and is explicitly excluded from the TTC computation chain. A four-state warning engine calibrated for laden-HCV braking dynamics incorporates a proactive brake-trigger boundary at 2.0 s TTC to compensate for actuation delay and sensor noise. When the inter-vehicle trajectory is non-closing, TTC is assigned the sentinel value $+\infty$, placing the decision logic in the `SAFE` state and ensuring both numerical stability and semantic clarity throughout the pipeline.

Across five evaluated scenarios, FleetEase achieved a mean detection-to-alert latency of 108 ms, a mean TTC estimation error of 0.10 s, an alert-classification accuracy of 97.3%, and a false-positive rate of 1.2%. Compared with a LiDAR-only baseline, it reduced missed detections by 41%. Compared with passenger-car thresholds, it improved correct critical-alert issuance by 23%. The simulation-first methodology and the modular sensor-abstraction architecture together provide a tractable validation and deployment path for HCV-oriented road safety in the Indian context. The openly documented pipeline serves as a reproducible reference for the Indian automotive research community.

Acknowledgements

The authors thank Dr. Deepa Nath and Dr. Apurva Naik for their continuous guidance, and the Department of Electrical and Electronics Engineering, MIT World Peace University, Pune, for providing computing resources and laboratory facilities.

Declarations

Funding. The authors received no external funding for this work.

Conflict of interest. The authors declare no conflicts of interest.

Data availability. Simulation data generated during this study are available from the corresponding author on reasonable request.

Code availability. Simulation scripts are available from the corresponding author on reasonable request.

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